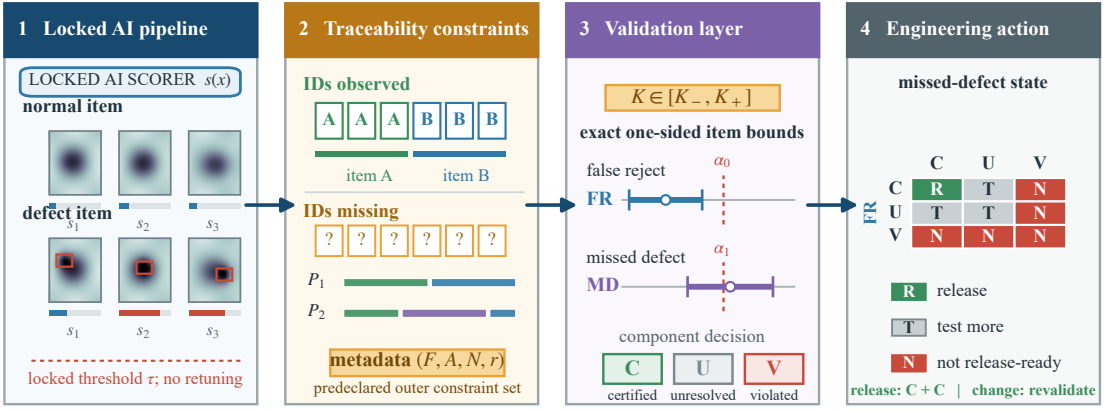

1 Locked AI pipeline
LOCKED AI SCORER $s(x)$
normal item
defect item
s_1 s_2 s_3
s_1 s_2 s_3
locked threshold τ; no retuning
2 Traceability constraints
IDs observed
A A A B B B
item A
item B
IDs missing
? ? ? ? ? ?
P_1
P_2
metadata (F, A, N, r)
predeclared outer constraint set
3 Validation layer
$K \in [K_-, K_+]$
exact one-sided item bounds
false reject
α_0
FR
missed defect
α_1
MD
component decision
C
U
V
certified
unresolved
violated
4 Engineering action
missed-defect state
C U V
FR
C R T N
U T T N
V N N N
R release
T test more
N not release-ready
release: C + C | change: revalidate